tum: it is transmuted to the internal form by the alchemy of an unsymmetrical state of stress. After transformation it may be accumulated within the system or removed to the surroundings. Just as the body-force is the rate of arrival of linear momentum from distant material surroundings, the body-couple $G_k$ is the rate of arrival of internal angular momentum from the same source; and in a similar analogy the net surface couple $C_{lk,j}$ is the rate of arrival, by diffusion, of internal angular momentum from contiguous material surroundings.

So far as we know, equation (5) was first given in complete form in 1959 by Dahler[1], who derived it as a general hydrodynamical equation in the context of a statistical mechanical theory of transport processes in fluids composed of polyatomic molecules. In his developments for classical and quantum cases, patterned after the methods of Irving and Kirkwood and of Irving and Zwanzig, a key step (as in the continuum development here) consists of distinguishing interactions of a system with distant from those with contiguous surroundings. Somewhat earlier Grad[2] had argued in favour of a version lacking only the body-couple term. Much earlier the Cosserats[3] had given another version lacking the accumulation term; long neglected, theirs has been revived in the modern theory of rheology (cf. refs. 4 and 5) and re-discovered in the theory of elasticity[6]. Other incomplete versions have appeared in various contexts, for example, the theories of liquid crystals and of what are called anisotropic fluids[7,8].

The derivation sketched here confirms that equation (5) is a basic law of continuum mechanics fully as general as Cauchy's. What are some of its consequences? In the first place, equations (2) and (5) indicate that moment of momentum and internal angular momentum are separately conserved only when the local state of stress is symmetrical. Therefore the usual proof of symmetry fails when there are couple stresses and accumulation of internal angular momentum, as Grad[2] has observed, as well as when there are body couples to be accounted for. This implies that whenever the adoption of a continuum model requires averaging over the centres of mass of sub-continuum units having rotational degrees of freedom, the stress-tensor is in general asymmetrical, its asymmetry stemming from the interaction of the centres-of-mass system with an interpenetrating twist or spin system[1]. An example is provided by Toupin's theory of the elastic dielectric[4]; a farther-reaching one is provided by the new Laval–LeCorre–Raman theory of elasticity[6,9,10]. Indeed, equation (5) is a means of joining that theory to extensive researches on nuclear and molecular spin systems in the solid state. It suggests, incidentally, a need for directing attention to the angular momentum aspect of magnetism in quantitative descriptions of macroscopic behaviour of not only solids but also fluids and less-well-defined forms of matter. As yet, in the physics of fluids there are few examples of antisymmetrical states of stress: predicted for polyatomic fluids in general by the statistical mechanical theory, they seem to have been first considered in detail only recently by Ericksen in a theory of certain non-Newtonian fluids[7,8]. The complete equation of internal angular momentum, which is basic to any full consideration of such states of stress, merits further attention in rheology, particularly as an aid to understanding non-linear behaviour. There might be interesting consequences for the hydrodynamic theory of helium II. The possible existence of asymmetrical stress seems to have been overlooked, and perhaps the peculiar system of spin-excitations in that liquid might be more conveniently described in terms of intrinsic angular momentum density and equation (5) than in terms of vorticity singularities. The equation does suggest the existence of unexplored effects of interchange between moment of momentum and internal angular momentum in macroscopic systems rotated at high angular velocities.

Another consequence of the angular momentum equation is the need it shows for discovering the constitutive relations giving the dependences of antisymmetrical stress, couple stress, and body-couples on a set of independent variables that completely describes the dynamical state. This need compels one to recognize that continuum models of real matter must be endowed with point structure, a notion that goes back to Duhem and the Cosserats[3–5]. This structure must then be characterized quantitatively. Further work on energy and entropy accounting, on constitutive relations, on certain applications to polar fluids, and on consequents for turbulent flow is in progress.

We are grateful to Prof. W. H. Schwartz, of Stanford University, for bringing the most recent papers by Ericksen to our attention.

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# APPLICATION OF PALÆOMAGNETISM TO GEOLOGICAL CORRELATION

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THE Monteregian Hills are formed from the erosional remnants of eight igneous plugs which intrude the Palæozoic sedimentary beds of south-eastern Quebec (Fig. 1). Work on these intrusive rocks[1] by a number of geologists has shown that they were derived from a common magmatic source of basic composition.

Suggestions have been made[2] to include in the Monteregian Hills series a number of alkaline intrusives which are located outside the immediate area

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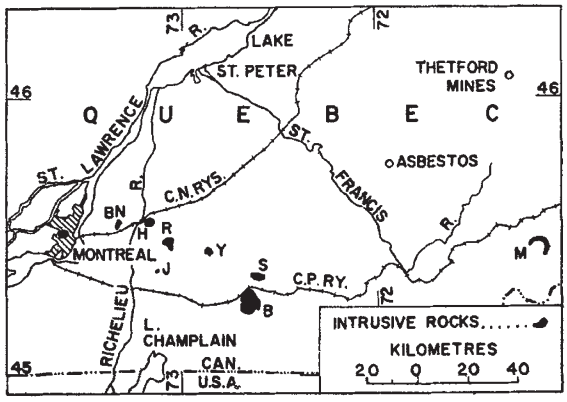



Fig. 1. Areal distribution of Monteregian hills (*BN, H, R, J, Y, S, B*) and Mount Megantic (*M*)

of the Hills. One of these bodies forms the collar of basic rocks on the outer rim of Mount Megantic, some seventy-five miles east of the main group. This body (labelled *M* in Fig. 1) was recently shown (Reid, A. M., unpublished M.Sc. thesis, University of Western Ontario, 1961) to bear petrological affinities with the Monteregian Hills series. The contemporaneity of the Mount Megantic rocks and the Monteregian Hills is suggested in this article on the basis of palæomagnetic data.

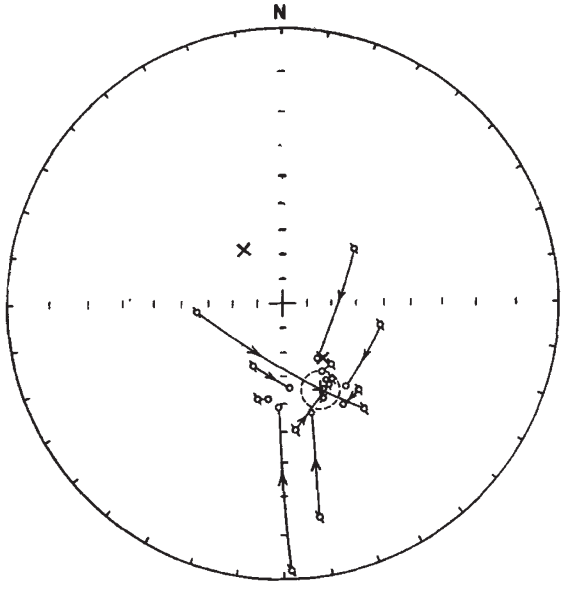



Fig. 2. Directions of magnetization of Mount Megantic rocks. $\phi$, original readings; ×, mean direction for other Monteregian hills; ○, after magnetic cleaning; ⊕, circle of confidence and mean direction for Mount Megantic rocks

A suite of twelve oriented specimens was collected from different sections of the basic part of Mount Megantic. For the sampling, a solar compass was used to eliminate the possible error that could be introduced by the interaction of the magnetic needle of an ordinary compass with the relatively strongly magnetic rocks dealt with. Two 1-in. cubes were cut from each specimen and their magnetization directions were measured. The mean direction of magnetization of the two cubes was taken as representative of the specimen. For the purpose of eliminating unstable components of magnetization in the rocks, the latter were then submitted to the action of a decreasing amplitude, alternating field in a magnetic-field-free space. The maximum initial field amplitude used was 250 œrsteds. This treatment improved the clustering of the magnetization directions a considerable amount, as illustrated in Fig. 2.

The mean direction of magnetization for the group is represented by a cross centring the dotted line circle of confidence, the radius of which was computed using Fisher's method.

The fact that the parameter $\hat{K}$ of the group is relatively high, coupled with the fact that the mean direction of magnetization is almost diametrically opposed to the direction of the present field of Earth at the collecting site, suggest strongly that the mean direction given in Fig. 1 is that of a stable component of magnetization in the rocks and that it reflects the attitude of the ambient magnetic field of Earth at the time the rocks cooled from magmatic temperatures.

In a previous study[3], I found that the mean directions of magnetization of the Monteregian Hills lie along an axis represented by two *X*'s in Fig. 2. The attitude of this axis was computed from the mean directions of magnetization of two groups of rocks which were respectively polarized downward in the north-west quadrant and upward in the south-east quadrant. Although the magnetization direction of the Mount Megantic rocks is almost parallel to this axis, it is more accurate, for the sake of geological correlation, to compare the pole positions, which may be inferred from the Mount Megantic and the Monteregian Hills palæomagnetic directions respectively. The reason for this is the remoteness of the two groups of collecting sites. These pole positions are represented stereographically in Fig. 3, and they are labelled '*m*' and '*M*' respectively. Their proximity and the overlapping of their respective ellipse of confidence strongly suggest that the Mount Megantic

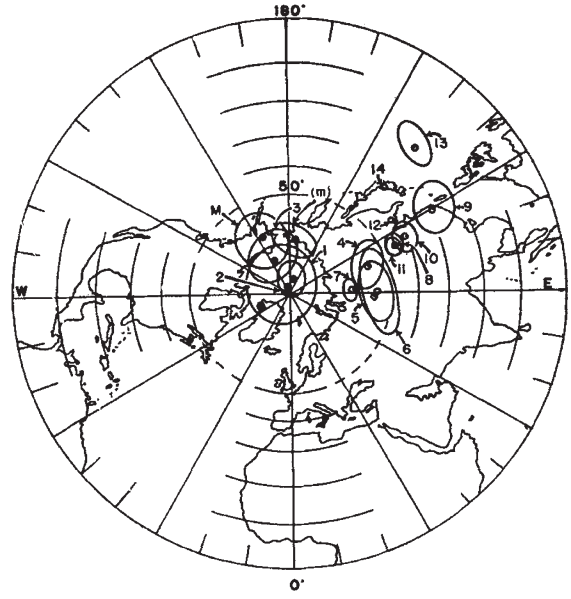



Fig. 3. Palæomagnetic pole positions inferred from North American rocks. *M*, Monteregian Hills, Mount Megantic excluded; *m*, Mount Megantic; 1, Tertiary rocks; 2, Pliocene and Miocene rocks; 3, Cretaceous rocks; 4, 5, 6, 7, Triassic rocks; 8, 9, 10, Permian rocks; 11, 12, Carboniferous rocks; 13, Silurian rocks

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Table 1. Directions of Magnetization of Basic Intrusive Rocks

$N$, Number of samples; $\theta$, radii of circles of confidence; $\hat{K}$, dispersion parameter; $D$, declination; $I$, inclination

| | $N$ | $\theta$ | $\hat{K}$ | $D$ | $I$ |
|---|---|---|---|---|---|
| Mount Megantic | 12 | 6·5 | 44 | 157 | −52 |
| Other Monteregian Hills | 10 | 18° | 7·9 | 340 | +57 |
| Other Monteregian Hills | 39 | 9° | 7·5 | 142 | −62 |

Table 2. Palæo Pole Positions

| | Latitude | Longitude |
|---|---|---|
| Monteregian Hills | 65° N. | 157° W. |
| Mount Megantic | 68·5 N. | 171·5 E. |
| Dakota sandstones* | 74 N. | 168 W. |

* Pole position as computed by me from directions of magnetization and collecting sites given by Runcorn (1956).

Table 3

| | Mineral in parent rock | Age in $10^6$ yr. | Laboratory |
|---|---|---|---|
| Mount Royal | Pyroxene in Tinguaite | 100 | M.I.T. |
| Mount Johnson | Biotite in Essexite | 110 | M.I.T. |
| Mount Brome | Biotite in Nordmarkite | 122 | G.S.C. |
| Mount Megantic | Biotite in Gabbro | 115 | G.S.C. |

Potassium/argon age determinations by Geological Survey of Canada (G.S.C.) and the Massachusetts Institute of Technology (M.I.T.).

basic rocks and the Monteregian Hills were intruded at the same time.

Pole positions derived from other North American rocks of known ages[4] are also plotted in Fig. 3. It appears from these data that a marked shift of the pole occurred between Cretaceous and Triassic times. It would seem, therefore, that pre- and post-Triassic rocks could be distinguished on the basis of their palæomagnetic directions. If this line of reasoning were legitimate, the '$M$' and '$m$' poles belong to the post-Triassic group and their proximity to the Cretaceous pole would suggest further a Cretaceous age for both the Mount Megantic and the Monteregian Hills rocks. This conjecture was recently supported by potassium/argon age determination data (Table 3) released by two independent laboratories on samples collected from three of the Monteregian Hills and from Mount Megantic. Thus, palæomagnetic and radiogenic data support independently the synchronous intrusions of these two rock units and place them in Cretaceous times.

I thank the Director of the Geological Survey of Canada for permission to publish this work.

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# DIFFERENTIAL SECONDARY MAGNETIZATION IN SOME BRITISH CAMBRIAN ROCKS

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TWELVE orientated rock specimens were collected from three horizons in the Hartshill Quartzites and the Hyolithus Limestones from two localities near Nuneaton. Geologically[1] the quartzites include true quartzites and ferruginous sandstones coloured red or pink, while the limestones consist of organic material in a fine-grained matrix of calcite, iron minerals and other material, the whole strongly red stained. The rocks dip to the south-west at about 40 deg.

Magnetic results were obtained for eleven rocks and are plotted on the equal area stereonet (Fig. 1) in their field orientations; the twelfth was too weak to give significant results with the existing magnetometer. Combining the vectors for the four specimens from the Hyolithus Limestones by taking the mean of the direction cosines and making a statistical estimate of the dispersion after Fisher[2] gives:

$$D\ 017°\quad I\ 23°\quad \alpha\ 16°\quad \varkappa\ 33$$

The following abbreviations are used: $D$, declination measured from true north; $I$, inclination to the horizontal, positive downwards; $\alpha$, semi-angle of the cone of confidence drawn at the 95 per cent probability-level; $\varkappa$, estimate of precision. In contrast, the mean direction obtained from the seven specimens from the quartzites is:

$$D\ 132°\quad I\ 64°\quad \alpha\ 25°\quad \varkappa\ 6$$

Correcting for the dip of the rocks moves both the mean directions towards the south-west by about 20 deg., so that when these directions are compared with those found in Cambrian rocks from Wales[3] the quartzites give a similar direction, but the mean vector for the limestones is about 90 deg. removed.

All the rocks were demagnetized in a.c. fields of up to 100 œrsteds using apparatus similar to that described by Creer[4]. The new vectors obtained on re-

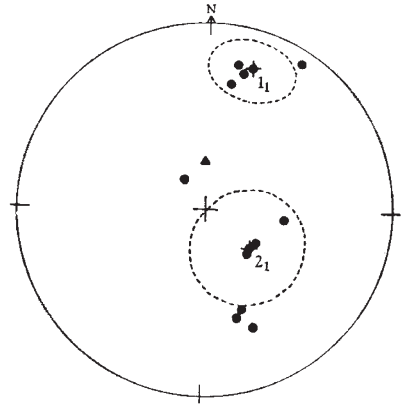



Fig. 1. Equal area stereonet showing the positions of the magnetic vectors for eleven rocks from the British Cambrian together with the group mean vectors. All vectors are in field orientation. ●, Vector plotting in lower hemisphere; $1_1$, mean vector for limestones; $1_2$, mean vector for limestones after a.c. washing; $2_1$, mean vector for quartzites; $2_2$, mean vector for quartzites after a.c. washing; $2_3$, mean vector for quartzites after a.c. washing and correction for dip; •, vector for Trias rocks; ■, vector for Cambrian rocks; ▲, direction of present dipole field. The broken lines show the circles of confidence drawn at the 95 per cent level

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